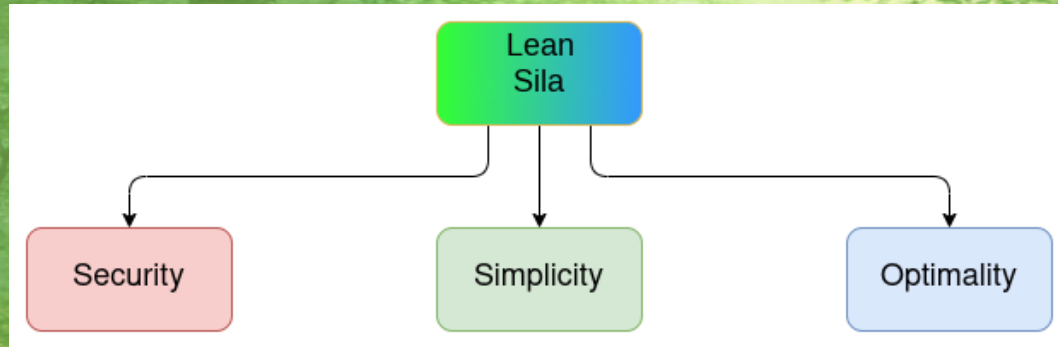


Lean Sila



Lean Sila: Sila for the long term



Why security?

- Sila L1 is the World Ledger
- Fast chains that break are a dime a dozen. Zero-downtime chains where your assets and records are safe are the rare thing
- The ability of everything else (L2, apps...) to function depends on L1 being safe

Why simplicity?

- Maximizes number of people who understand the protocol and feel empowered to contribute
- Forcing function toward ossification
- Feeds into security

Rsil correctly implemented the SIP-7702 specification, checking if an account is "empty" (per SIP-161's definition), whereas gsil's implementation of the SIP-7702 pre-execution checks checked if the account existed within the state. This discrepancy cannot affect Sila SilaMainnet, because empty accounts are unable to be persisted within the state following SIP-161 (included within the Spurious Dragon hardfork), and all known empty accounts were cleared from the state with <https://silascan.io/tx/0xf955834bfa097458a9cf6b719705a443d32e7f43f20b9b0294098c205b4bcc3d>.

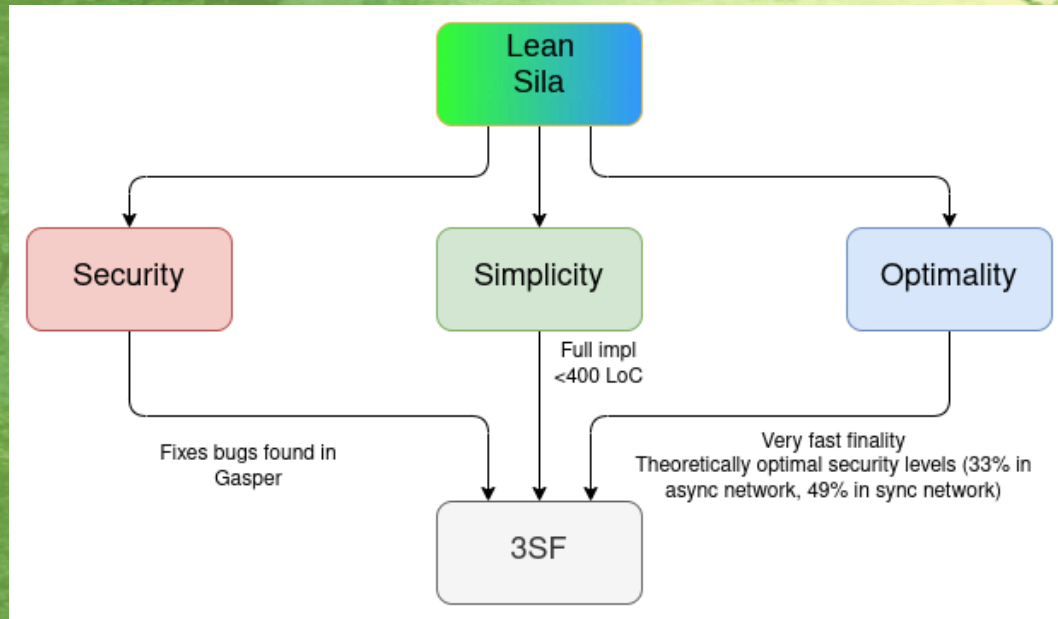
Why optimality?

- We want to scale and minimize latency
- Making Sila to be as efficient as possible is the only way to maximize scale and minimize latency without tradeoffs
- Feeds into ossification (if it's optimal, no reason to change it again)

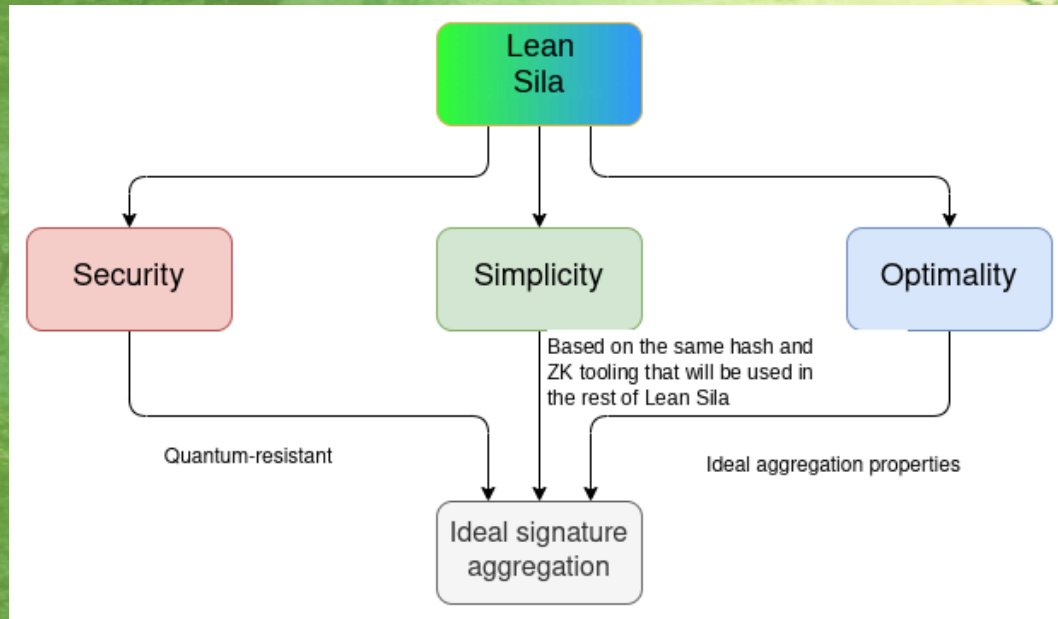
Why now?

- Consensus research is relatively mature
- ZK is more well-understood
- We have learned a lot from our mistakes
- The risk that we will miss a 10x gain from making the wrong choice is *much* lower than before

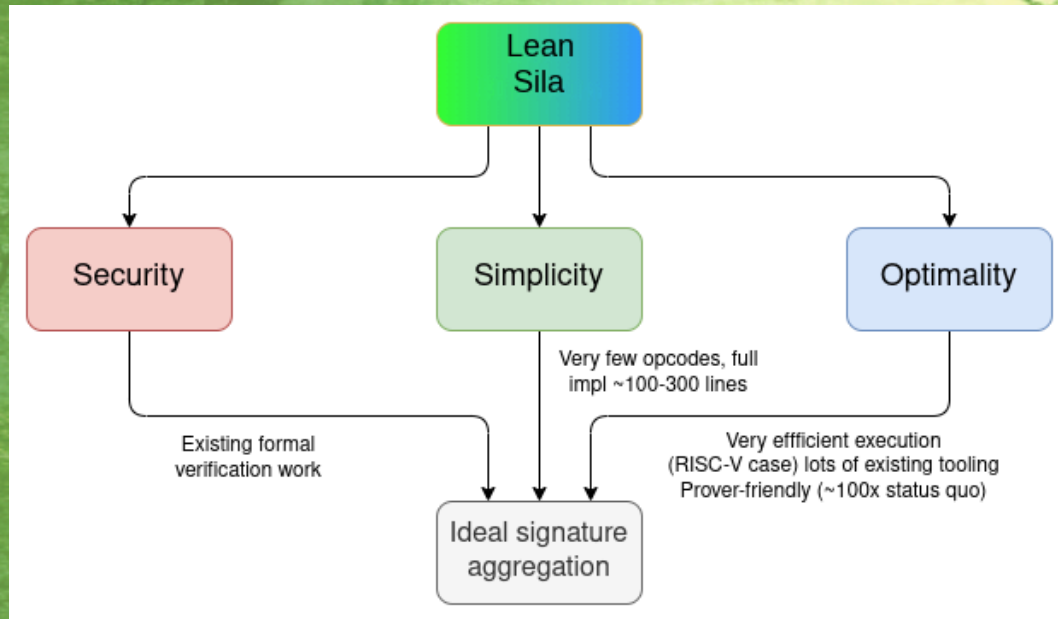
Example: 3SF



Example: ideal signature aggregation



Example: ZK-VM (RISC-V or Cairo-like)



Example: Data layer

